

M001 · Diamond Structure (Final Project) — English Worksheet

Topic: Diamond Structure (Final Project) · **Course:** Pyramid Problems · **Time:** about 60 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```

set max_size to 9

# Bottom half
set size to 1
while size <= max_size:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
    
```

What shape is the bottom half? List the sizes from bottom to top.


```

set max_size to 9

# Top half
set size to max_size - 2
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
    
```

What shape is the top half? List the sizes from bottom to top.


```
set max_size to 9

# Bottom half (1 → 9)
[grow from 1 up to 9]
# Top half (7 → 1)
[shrink from 7 down to 1]
```

Put together, what does the agent build? Why does the top half start at 7 (not 9)?

2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

Bug A — The agent is supposed to build a diamond: bottom half grows $1 \rightarrow 9$, top half shrinks $7 \rightarrow 1$. The two halves should meet at the widest layer.

```
set max_size to 9

set size to 1
while size <= max_size:
    [build layer]
    move up by 1
    set size to size + 2

set size to max_size
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

Hint: the top half repeats the widest layer. Fix it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to use **one variable** (`max_size`) to control the entire diamond size.

```

# Bottom half
set size to 1
while size <= 9:
    [build layer]
    move up by 1
    set size to size + 2

# Top half
set size to 7
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2

```

Hint: there are two hard-coded numbers (9, 7). Rewrite so changing `max_size` changes both.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The diamond's two halves should be **vertically stacked** (top half sits on the widest layer of the bottom half).

```
set max_size to 9

set size to 1
while size <= max_size:
    [build layer]
    set size to size + 2

set size to max_size - 2
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

Hint: the bottom half never moves up. Fix it.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build your diamond using **one** `max_size` **variable**, then try `max_size = 9` and a different value (e.g. 5 or 11) to see how it changes.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: ____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit

Send this worksheet + your video to teacher on KakaoTalk.